Jake Amaral

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Education

University of Florida Aug 2024 – May 2028
*B.S. in **Computer Science** and Minor in **Economics*** *Gainesville, FL*

- **GPA:** 3.52
- **Relevant Coursework:** Data Structures & Algorithms, Prog in C++, Prog in Python, Linear Algebra
- **Extracurricular:** Kappa Sigma Fraternity, Association for Computing Machinery

Technical Skills

Languages: Python, C++, C#, SQL (PostgreSQL), HTML, TypeScript
Frameworks: Node.js, Next.js, React, Tailwind, shadcn, Drizzle ORM, AWS SAM
Developer Tools: Git/GitHub, Vercel, Supabase, Clerk, Unity, Google OAuth, AWS, Docker, React Native

Experience

Design Team – Product Manager Sep 2026 – Present
UF Association for Computing Machinery *Gainesville, FL*

- Developing an event check-in platform for **UF's largest pre-professional computing org (2,300+ members)**, scaling to **thousands** of students across partner orgs (Next.js, TypeScript, Supabase/PostgreSQL)
- Building a cross-platform iOS/Android app with Expo (React Native) on a shared Supabase backend secured by Clerk JWT auth and row-level security
- Leading mobile team onboarding as PM: scope Jira tickets, assign work by developer strengths, and guide each member through shipping a PR via code review and CI

Software Engineering Intern May 2026 – Jul 2026
Safe Cities NPO *Cape Town, South Africa*

- Led a 4-person team to build and ship a full-stack platform now used by **50+** nonprofit staff, using Next.js, TypeScript, SQL (PostgreSQL), and Drizzle ORM
- Enforced role-based access for every user by implementing Clerk authentication with Google OAuth 2.0 and Calendar API integrations
- Delivered a production-ready application from ambiguous requirements by interviewing stakeholders and iterating designs directly with Safe Cities staff
- Gained a broader perspective on global software development and cross-cultural collaboration through **direct cultural immersion**

Projects

DormU | *Next.js, TypeScript, React Three Fiber, Gemini API, Meta SAM 3D, Twinly API, Vercel* Oct 2026

- Won **2nd Place, Best Use of the Twinly API** at Gator Hack V (UF AI Days) with a 3D dorm planner that lets incoming UF students place what they plan to bring in real-scale models of 24 rooms. Built the AI pipeline that turns one photo into a 3D model (Gemini detection, background removal, Meta SAM 3D), while a rules engine checks against official housing rules. Shipped this project with **529 unit** and **120 end-to-end tests** in 48 hours with a five-person, five-major team.

Spatial Math | *C#, Unity, Python, AWS (Bedrock, Lambda, SAM), Docker* Apr 2026 – Present

- Built a real-time 3D graphing calculator in Unity that compiles typed equations into functions and renders mapped surfaces of **3,700+ vertices**, with tangent planes computed by differentiation. Deployed a **serverless AWS backend** (SAM, Lambda, Docker) that pairs the C# engine with an LLM on Amazon Bedrock that explains results.

Enterprise Decision Making (EDM) AI Hackathon Entry | *IBM watsonx Orchestrate* Feb 2026

- Designed a workflow of **6 agents** in IBM's watsonx AI Agent Builder to support business decision-making and governance at scale. Built specialized extraction agents that surfaced inconsistencies and policy concerns hidden across **60+** synthetic data points in a structured JSON dataset. Kept humans in charge of final outcomes with an auditable review process, improving decision consistency and transparency.

Languages & Interests

Languages: English (Native), Spanish (Intermediate High), Portuguese (Basic)
Hobbies: Baseball, Electronic Music Production, Vibe-Coding, Golf, Martial Arts, Traveling